

Advanced Dynare

Chapter VII *Identification*

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Outline

1. The problem
2. Local Identification
3. Spectrum
4. Dynare Implementation
5. Identification Strength
6. Advanced Mode

Motivation

- We estimated parameters θ by evaluating the posterior $\mathcal{L}(\theta|y^T)$
- But we never formally verified whether all elements of θ are properly identified
- Parameter identification
 - is a regularity condition
 - can and should be assessed before taking your model to data
 - is a model property

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Simple example

- Consider simple monetary model without price rigidities and real shocks
- Real interest r is fixed at steady-state level
- Inflation π_t is determined via Fisher equation and central bank reaction function that determines nominal interest rate i_t :

$$i_t = r + E_t\pi_{t+1} \quad (1)$$

$$i_t = r + \phi\pi_t + \nu_t \quad (2)$$

- The two equations imply

$$E_t\pi_{t+1} = \phi\pi_t + \nu_t \quad (3)$$

- We need $\phi > 1$ (**Taylor-principle**) for Blanchard-Kahn conditions to be satisfied
- Otherwise: indeterminacy

Solution and estimation

- Assume error term is serially correlated (interest smoothing, policy mistakes, etc.)

$$\nu_t = \rho\nu_{t-1} + \varepsilon_t \quad (4)$$

- Iterate equation (3) forward:

$$\begin{aligned}\pi_t &= -\sum_{j=0}^{\infty} \left(\frac{1}{\phi}\right)^{j+1} E_t \nu_{t+j} = -\frac{1}{\phi - \rho} \nu_t \\ &= -\frac{1}{\phi - \rho} (\rho\nu_{t-1} + \varepsilon_t) = \rho\pi_{t-1} - \frac{\varepsilon_t}{\phi - \rho} = \rho\pi_{t-1} + w_t, \end{aligned} \quad (5)$$

where w_t is the new error term (ε_t and ν_t not observed)

- In equilibrium, the nominal interest rate is given by

$$i_t = r + E_t \pi_{t+1} = r + \rho\pi_t \quad (6)$$

- Regression of (2) will not estimate ϕ , but autocorrelation ρ
- Reason: inflation and policy shock ν_t are perfectly correlated to assure unique stable solution
- Central bank reaction describes off-equilibrium behavior, it cannot be identified from equilibrium observations!

The likelihood function

- Remember: Likelihood function is a **sufficient statistic**
- The likelihood function in this case is given by

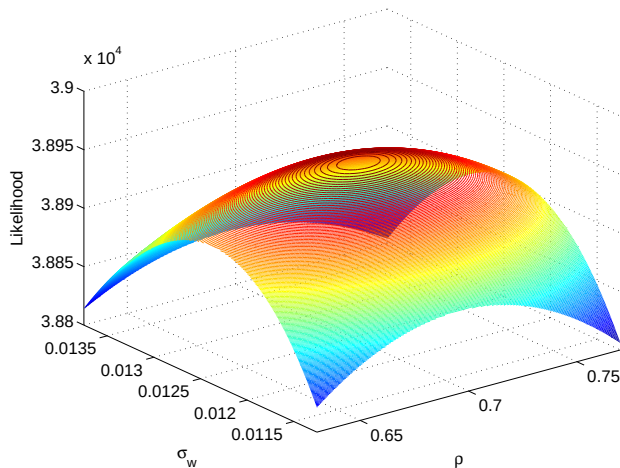
$$\log \left(f \left(\pi_t | \pi^{t-1} \right) \right) = -\frac{1}{2} \log (2\pi) - \frac{1}{2} \log \left(\sigma_{w_t}^2 \right) - \frac{1}{2} \frac{(\pi_t - \rho \pi_{t-1})^2}{\sigma_{w_t}^2} \quad (7)$$

where

$$\sigma_{w_t}^2 = \left(\frac{\sigma_\varepsilon}{\phi - \rho} \right)^2 \quad (8)$$

- ρ and σ_{w_t} enter the likelihood function independently of each other and can be estimated

Likelihood plot for ρ and σ_{w_t}



The likelihood function shows a clear peak in ρ and σ_{w_t}

Estimating the components of σ_{w_t}

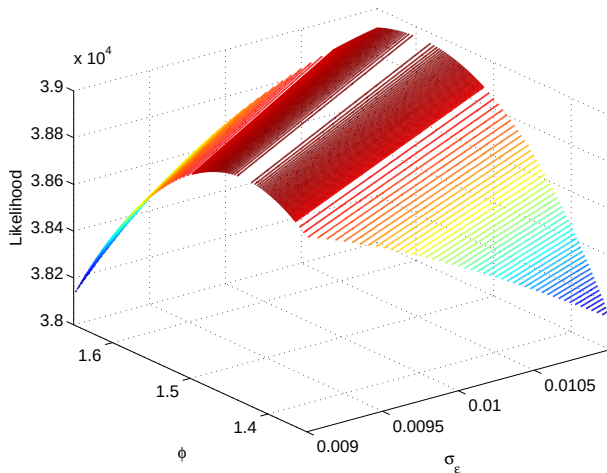
- But: σ_ε and ϕ enter multiplicatively and cannot be disentangled

$$\begin{aligned}\frac{\partial \log (f\left(\pi_t \mid \pi^{t-1}\right))}{\partial \sigma_\varepsilon} &= \frac{\partial \log (f\left(\pi_t \mid \pi^{t-1}\right))}{\partial \sigma_{w_t}} \frac{\partial \sigma_{w_t}}{\partial \sigma_\varepsilon} \\ &= \frac{\partial \log (f\left(\pi_t \mid \pi^{t-1}\right))}{\partial \sigma_{w_t}} \frac{1}{\phi-\rho}\end{aligned}\tag{9}$$

$$\begin{aligned}\frac{\partial \log (f\left(\pi_t \mid \pi^{t-1}\right))}{\partial \phi} &= \frac{\partial \log (f\left(\pi_t \mid \pi^{t-1}\right))}{\partial \sigma_{w_t}} \frac{\partial \sigma_{w_t}}{\partial \phi} \\ &= \frac{\partial \log (f\left(\pi_t \mid \pi^{t-1}\right))}{\partial \sigma_{w_t}} \frac{\sigma_\varepsilon}{(\phi-\rho)^2}(-1) \\ &\stackrel{(9)}{=} -\frac{\sigma_\varepsilon}{\phi-\rho} \frac{\partial \log (f\left(\pi_t \mid \pi^{t-1}\right))}{\partial \sigma_\varepsilon}\end{aligned}\tag{10}$$

- The partial derivatives are linearly dependent, and the Jacobian is thus singular

Likelihood plot for σ_ε and ϕ



The likelihood function has no unique maximum in σ_ε and ϕ , only a range of maxima

Economic equations

- Consider the purely forward-looking three equation NK model

$$r_t = \psi \pi_t + \varepsilon_t^M \quad (11)$$

$$x_t = E_t x_{t+1} - \frac{1}{\tau} (r_t - E_t \pi_{t+1}) + \varepsilon_t^D \quad (12)$$

$$\pi_t = \beta E_t \pi_{t+1} + \kappa x_t + \varepsilon_t^S \quad (13)$$

or in matrix form

$$\begin{pmatrix} 1 & 0 & -\psi \\ \frac{1}{\tau} & 1 & 0 \\ 0 & -\kappa & 1 \end{pmatrix} \begin{pmatrix} r_t \\ x_t \\ \pi_t \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & \frac{1}{\tau} \\ 0 & 0 & \beta \end{pmatrix} \begin{pmatrix} E_t r_{t+1} \\ E_t x_{t+1} \\ E_t \pi_{t+1} \end{pmatrix} + \begin{pmatrix} \varepsilon_t^M \\ \varepsilon_t^D \\ \varepsilon_t^S \end{pmatrix} \quad (14)$$

- If $A_0^{-1} A_1$ is stable, system $A_0 y_t = A_1 E_t y_{t+1} + \varepsilon_t$ can be solved forward:

$$y_t = A_0^{-1} A_1 E_t y_{t+1} + A_0^{-1} \varepsilon_t = \sum_{j=0}^{\infty} (A_0^{-1} A_1)^j A_0^{-1} E_t \varepsilon_{t+j} = A_0^{-1} \varepsilon_t \quad (15)$$

Purely forward-looking model

- Solution

$$\begin{pmatrix} r_t \\ x_t \\ \pi_t \end{pmatrix} = \frac{1}{\frac{\kappa\psi}{\tau} + 1} \begin{pmatrix} 1 & \kappa\psi & \psi \\ -\frac{1}{\tau} & 1 & -\frac{\psi}{\tau} \\ -\frac{\kappa}{\tau} & \kappa & 1 \end{pmatrix} \begin{pmatrix} \varepsilon_t^M \\ \varepsilon_t^D \\ \varepsilon_t^S \end{pmatrix} \quad (16)$$

shows several issues:

- β does not enter solution and therefore any objective like the likelihood
- Identification depends on observables, e.g., when observing only x_t :
 - τ and κ are pairwise collinear
 - κ and ψ are pairwise collinear
→ we would need to fix e.g., ψ

Identification problems in theory

1. **Observational Equivalence:** mapping between structural parameters and objective function has no unique maximum

⇒ structural models with potentially different economic interpretations may be indistinguishable
2. **Under-identification:** objective function is independent of certain structural parameters, e.g. because they disappear from rational expectations solution

⇒ **partial identification** with two structural parameters entering objective function only proportionally, making them separately unrecoverable, is a special case
3. **weak identification:** parameter theoretically identified, but curvature may be small in certain regions of the parameter space

Identification problems in practice

- Lack of identification leads to wrong conclusions from calibration, estimation, and inference
- Many caveats due to identifiability issues and/or an unfortunate choice of observables
 - difficult to maximize likelihood/posterior or minimize some (moment) objective function
→ singular Hessian
 - estimators often lie on the boundary of theoretically admissible space
 - Gaussian asymptotic theory yields poor approximations
- Weak identification is likely a more serious concern for applied researchers

Is identification even a problem for Bayesians?

- Won't we just get our prior as the posterior if a parameter is not identified?
- Unfortunately, the general answer is no
- Koop et al. (2013): determinacy conditions in DSGE models cause identified parameters to have spill-over on non-identified ones, even if independent priors were assumed
- Forward-looking example: eigenvalues of $A_0^{-1}A_1$ needed to lie within the unit circle
→ implicit joint prior space is smaller than product of explicit individual ones ("parameter space not variation-free")
- Consequence: posterior distribution differs from the prior even when the parameter is unidentified
→ apparent learning (changing posterior) need not be informative about identification

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State space system: transition equation

- Is there a systematic way to detect such issues for elements of θ ?
- Consider generic log-linearized model in state-space form where $\hat{x}_t = \log(x_t) - \log(x)$

$$\hat{x}_{t+1} = F(\theta)\hat{x}_t + w_{t+1}, \quad w_{t+1} \stackrel{iid}{\sim} \mathcal{N}(0, Q(\theta)) \quad (17)$$

with

$$Q = CC'$$

- Define the vector

$$\tau = [\tau'_x, \tau'_F, \tau'_\Omega] \quad (18)$$

as the collection of non-constant elements of $\log(x)$, $vec(F)$, and $vech(Q)$, respectively, that depend on θ

- Drops, e.g., steady states that are always zero or calibrated values
- The unconditional first and second moments are given by:

$$E(\hat{x}_t) = 0$$

$$E(\hat{x}_t \hat{x}_t') \equiv \Sigma_x(0) = F\Sigma_x F' + Q$$

State space system: observation equation

- The observables are given by

$$y_t = H(\theta)x + H(\theta)\hat{x}_t + \nu_t, \nu_t \stackrel{iid}{\sim} N(0, R(\theta)) \quad (19)$$

- In general, only observing y_t would be insufficient to fully characterize the distribution of x_t
- Fortunately, our model implies restrictions through θ

The likelihood function

- The unconditional first and second moments of the observables y_t are given by:

$$E(y_t) \equiv \mu_Y = H(\theta) x$$
$$\text{cov}(y_{t+i}, y_t) \equiv \Sigma_Y(i) = \begin{cases} C \Sigma_x(0) C' & \text{if } i = 0 \\ C F^i \Sigma_x(0) C' & \text{if } i > 0 \end{cases}$$

- Stack the unique elements of $\Sigma_Y(i)$, $i = \{0, \dots, T-1\}$ into a vector:

$$\sigma_Y = [\text{vech}(\Sigma_Y(0)), \text{vec}(\Sigma_Y(1)), \dots, \text{vec}(\Sigma_Y(T-1))] \quad (20)$$

and define the vector of first and second moments as

$$m_T = [\mu_Y; \sigma_Y] \quad (21)$$

Definitions

Definition 1 (Global Identification)

Let $\theta \subseteq \Theta \subset \mathcal{R}^k$ be the parameter vector of interest and suppose inference about it is made based on T observations of a random vector y with known joint probability density function $f(Y^T|\theta)$. A point $\theta_0 \in \Theta$ is **globally identified** if

$$f(Y^T|\tilde{\theta}) = f(Y^T|\theta_0) \text{ with probability 1} \Rightarrow \tilde{\theta} = \theta_0 \quad (22)$$

for any $\tilde{\theta} \in \Theta$

Definition 2 (Local Identification)

If Equation (22) only holds in an open neighborhood of θ_0 , then θ_0 is **locally identified**

Which moments?

- Typically, the distribution is unknown or assumed to be normal
- People often use the first two moments
- Higher order moments might provide additional information (see e.g. Fernández-Villaverde and Rubio-Ramírez 2007)
- Thus, identification based on mean and (co)-variances is sufficient, but not necessary

Global identification condition

Theorem 3 (Global Identification Condition)

Suppose our model in state-space form with θ_0 is the data generating process for Y^T . Then θ_0 is **globally identified** if

$$m_T(\tilde{\theta}) = m_T(\theta_0) \Leftrightarrow \tilde{\theta} = \theta_0 \quad (23)$$

for any $\tilde{\theta} \in \Theta$ (and correspondingly for local identification).

- Requires mapping from population moments m_T to θ to be unique
 - If not satisfied, there exists parameter values that imply the same joint distribution
- ⇒ Even with infinitely many observations, the true value could not be identified
- Checking global identification is hard ⇒ check local identification for relevant parameter range

Local identification condition

Theorem 4 (Local Identification Condition)

Suppose m_T is continuously differentiable function of θ . Then θ_0 is locally identifiable if the Jacobian matrix

$$J(q) \equiv \frac{\partial m_q}{\partial \theta'} \quad (24)$$

has full column rank at θ_0 for $q \leq T$

Proof.

Follows from implicit function theorem. □

- We need at least as many moments as parameters
- For sufficient and necessary conditions see Iskrev (2010)

Identification conditions: intuition

- Identification requires unique (injective) mapping from structural parameters to population moments
- Check injectivity by looking at rank of Jacobian matrix
- One would have to check this condition everywhere $\Rightarrow$ impossible
- Nevertheless helpful in detecting observational equivalence (column of zeros) and under-identification (linear dependence of columns)

Computing the Jacobian

- Numerical differentiation and checking for singularity of the numerical Jacobian is prone to numerical errors when functions are nonlinear

⇒ Use analytical derivatives

- Using the chain rule, the Jacobian in equation (24) can be written as

$$J(T) \equiv \underbrace{\frac{\partial m_T}{\partial \tau'}}_{J_1(T)} \underbrace{\frac{\partial \tau}{\partial \theta'}}_{J_2(T)} \quad (25)$$

- Iskrev (2010) details how the two parts of the Jacobian can be computed, but we take Dynare's implementation
- Full rank is often achieved for $q \ll T$: Dynare starts at $q = 5$ and increases it if necessary

Decomposing the Jacobian

$$J(T) \equiv \underbrace{\frac{\partial m_T}{\partial \tau'}}_{J_1(T)} \underbrace{\frac{\partial \tau}{\partial \theta'}}_{J_2(T)} \quad (25)$$

- $J_2(T)$ shows how the parameters θ affect the non-constant model solution parts τ
- $J_1(T)$ shows how the model solution maps into the observed data moments

Corollary 5

The point θ_0 is locally identifiable only if $J_2(T)$ has full rank

- Necessary condition as parameters only affect distribution of observables through their effect on model solution
- It is not sufficient unless all states are observed
- $J_1(T)$ provides information on whether a theoretically identifiable parameter can be identified using the observables contained in y

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Diagnostics based on spectrum

- State space solution also allows computing theoretical spectral density $\mathcal{S}_{2,x}(\omega)$ and $\mathcal{S}_{2,y}(\omega)$ for $\omega \in [-\pi, \pi]$
- Qu and Tkachenko (2012) show that identification implies unique (injective) mapping from structural parameters to population mean and spectral density
→ check injectivity by looking at rank of Jacobian matrix
- Rank condition: Jacobian of theoretical mean and spectrum of observables w.r.t structural parameters has full rank
- Again, helpful in detecting observational equivalence (columns of zeros and linear dependence between parameters or try all combinations of sets of parameters);
- Gram matrix structure numerically facilitates rank computations

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Dynare output for the Cochrane toy model

cochrane_toy.mod

REDUCED-FORM:

All parameters are identified in the Jacobian of steady state and reduced-form solution matrices

MOMENTS (ISKREV, 2010):

!!!WARNING!!!

The rank of J (Jacobian of first two moments) is deficient!

[sig_nu,phi] are PAIRWISE collinear!

- All parameters are identified in the model
- However, without observing ν_t , the elements of σ_w cannot be disentangled

Dynare output for the forward-looking model

forward_looking_varobs_x.mod

REDUCED-FORM:

!!!WARNING!!!

The rank of Tau (Jacobian of steady state and reduced-form solution matrices) is deficient!

BETA is not identified!

SPECTRUM (QU AND TKACHENKO, 2012):

!!!WARNING!!!

The rank of Gbar (Jacobian of mean and spectrum) is deficient!

BETA is not identified!

[TAU,PSI] are PAIRWISE collinear!

[KAPPA,PSI] are PAIRWISE collinear!

- β is not identified
- Parameter collinearity is detected

Dynare output for our RBC model

RBC_bayesian.mod

REDUCED-FORM:

All parameters are identified in the Jacobian of steady state and reduced-form solution matrices (rank(Tau) i

MINIMAL SYSTEM (KOMUNJER AND NG, 2011):

All parameters are identified in the Jacobian of steady state and minimal system (rank(Deltabar) is full with

SPECTRUM (QU AND TKACHENKO, 2012):

!!!WARNING!!!

The rank of Gbar (Jacobian of mean and spectrum) is deficient!

rhoz is not identified!

MOMENTS (ISKREV, 2010):

All parameters are identified in the Jacobian of first two moments (rank(J) is full with tol = robust).

- All estimated parameters are identified in the model and using our observables

identification-command

- Triggers the local identification tests and has two modes of operation:
 1. Point identification check (default) at `parameter_set` (default `prior_mean`)
 2. Monte Carlo mode (e.g. `prior_mc=1000`)
- By default all structural model parameters and all `stderr` parameters are checked
- Other parameter sets to be checked can be selected with `estimated_params` block

A warning

- Identification checks typically require symbolic derivatives to achieve sufficient precision
- Dynare's symbolic preprocessor interprets and implements the model definitions as expressed in the Dynare file
 - will not reflect all parameter definitions hidden e.g. in a custom `_steadystate.m` file
 - use `steady_state_model` block instead
- Do not declare auxiliary parameters, rather use model-local variables ('#' syntax in the model block) to declare endogenous parameters

Tracking singularities

- When Jacobians are rank deficient, code tries to diagnose subset of parameters responsible:
 - ranks are computed using the singular value decomposition
 - for columns of zeros in Jacobian, associated parameter is printed on command window
 - compute pairwise- and multi-correlation coefficients for each column of Jacobian
→ print parameter combinations with correlation coefficients equal to unity
- Alternatively: a brute-force approach to check rank conditions for all possible combinations can be triggered (`checks_via_subsets=1`)

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Information Matrix revisited

- The precision of the parameter estimates at the mode was given (asymptotically) by the inverse of the **Fisher Information Matrix**

$$I(\theta) = E \left[\left(\frac{\partial \log(p(Y|\theta))}{\partial \theta'} \right)' \left(\frac{\partial \log(p(Y|\theta))}{\partial \theta'} \right) \right] \quad (26)$$

- Typically, non-singularity of this matrix is sufficient for local identification (for details see Iskrev 2011)
- Can be decomposed into a variance matrix Δ and a correlation matrix $\tilde{I}(\theta)$

$$I(\theta) = \Delta^{\frac{1}{2}} \tilde{I}(\theta) \Delta^{\frac{1}{2}} \quad (27)$$

where

$$\Delta = \text{diag}(I(\theta))$$

Information Matrix and weak identification

$$I(\theta) = \Delta^{\frac{1}{2}} \tilde{I}(\theta) \Delta^{\frac{1}{2}} \quad (27)$$

- Here, we can again see our two reasons for non-identifiability:

1. The likelihood does not change when parameter i changes, i.e.

$$\Delta_i = 0 \quad (28)$$

2. The effect on the likelihood is offset due to perfect correlation, i.e.

$$\rho_i \equiv \text{corr} \left(\frac{\partial \log(p(Y|\theta))}{\partial \theta_i}, \frac{\partial \log(p(Y|\theta))}{\partial \theta_{-i}} \right) = 1 \quad (29)$$

- This suggests that we can use this curvature information also for **weak identification** as in that case $\Delta_i \approx 0$ and/or $\rho_i \approx 1$

Unnormalized identification strength

- Identification strength can be measured using

$$s_i(\theta) = \sqrt{\Delta_i (1 - \rho_i^2)} \quad (30)$$

- If $I(\theta)$ is not singular, $1/s_i(\theta)$ is equal to the square root of the i th diagonal element of $(I(\theta))^{-1}$, i.e.

$$s_i(\theta) = \sqrt{\frac{1}{((I(\theta))^{-1})_{(ii)}}} \quad (31)$$

- As this measure only uses the population objective function (expected log-likelihood), it is an a-priori measure
- At the same time, it can be interpreted as the lower bound on the estimation uncertainty in an unbiased finite sample estimator

Normalized identification strength

- Problem: measure is percentage change in likelihood for unit change in parameter $\rightarrow$ not unit free
- Solution: multiply with value of θ_i to obtain elasticity measure:

$$s_i^r(\theta) = \sqrt{\theta_i^2 \Delta_i} \times \sqrt{(1 - \rho_i^2)} \quad (32)$$

where the first part is the elasticity of the likelihood function w.r.t. θ_i , keeping all other parameters constant

Dynare implementation

- The strength of identification for parameter θ_i is defined through a renormalized version of the curvature as

$$s_i^{dyn} = \theta_i \times s_i(\theta) \stackrel{(31)}{=} \sqrt{\frac{\theta_i^2}{(I(\theta)^{-1})_{(i,i)}}} \quad [STRENGTH] \quad (33)$$

- Can be interpreted as an “a-priori t-test”
- Taking square of θ assures it to be positive
- The **sensitivity component** contained in this measure (as opposed to the **correlation component**) is defined as

$$\tilde{\Delta}_i = \sqrt{\theta_i^2 \Delta_i} = \sqrt{\theta_i^2 I(\theta)_{(i,i)}} \quad [SENSITIVITY] \quad (34)$$

- Identification is often checked over the prior region, e.g. at the prior mean

Dynare implementation with alternative normalization

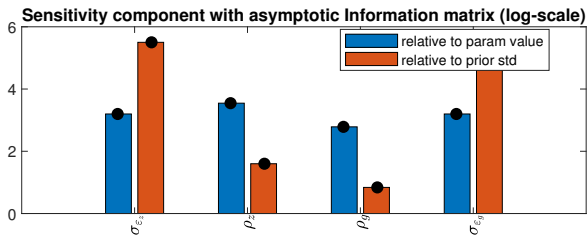
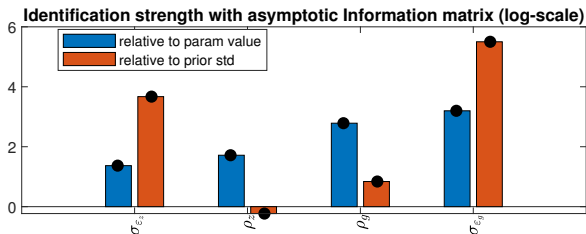
- Because some parameters may have a prior mean of zero, an alternative normalization uses the prior standard deviation $\sigma(\theta_i)$:

$$s_i = \sqrt{\frac{(\sigma(\theta_i))^2}{(I(\theta)^{-1})_{(i,i)}}} \quad [STRENGTH] \quad (35)$$

$$\tilde{\Delta}_i = \sqrt{(\sigma(\theta_i))^2 I(\theta)_{(i,i)}} \quad [SENSITIVITY] \quad (36)$$

Dynare Output: identification strength and sensitivity

RBC_bayesian.mod



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The advanced=1 option

- Additional analysis of the LRE form (steady-state and dynamic model derivatives) for all model variables
- Different Sensitivity Measure based on norm of columns of Jacobians
- Analysis of identification patterns based on regressions and Singular Value Decompositions

A different sensitivity measure

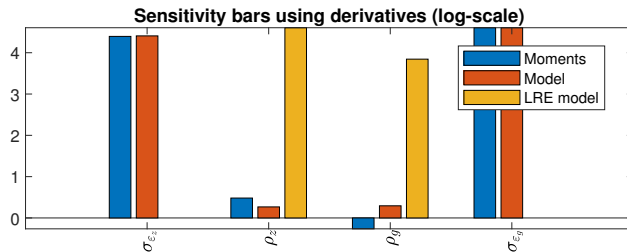
- Another sensitivity measure is how changes in the elements of the parameter vector θ impact the model moments
- Can be measured locally using the Jacobian, where the entry $\partial m_j / \partial \theta_i$ needs to be normalized
- Can be done by multiplying with the ratio of standard deviations

$$\text{std}(\theta_i) / \text{std}(m_j) \quad (37)$$

- Accounts for different parameter uncertainty by ascribing more importance to more variable parameters (effectively normalizing across parameters i)
- Normalization with standard deviation $\text{std}(m_j)$ allows for comparing the impact of parameter i on differently volatile moments (normalizing across moments j).
- Norm of the columns of the standardized Jacobian yields single aggregate sensitivity measure over all moments j for each parameter i .

Dynare Output: sensitivity measure components

RBC_bayesian.mod



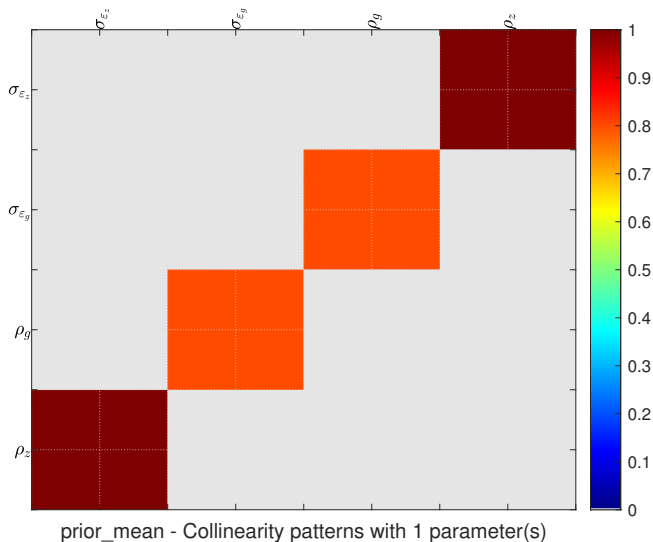
- The respective Jacobian matrices refer to
 1. the moments matrix ($\partial m_T / \partial \theta'$), indicating how well a parameter can be identified due the strength of its impact on the observed moments
 2. the model solution matrices ($\partial \tau / \partial \theta'$), indicating how well a parameter could in principle be identified if all state variables were observed
 3. the Linear Rational Expectations model ($\partial \gamma / \partial \theta'$), indicating trivial cases of non-identifiability due to e.g. some parameters always showing up as a product in the model equations

Correlation Component

- Even if the likelihood itself is sensitive, there might be (perfect) collinearity
- Need to look at correlation component
- This analysis is conducted via brute force: For each single parameter, a set of regressions is run of the column of the Jacobian corresponding to the parameter in the row on all possible combinations of other Jacobian columns
- Aim: finding the column (and thus parameter) combination with the highest R^2

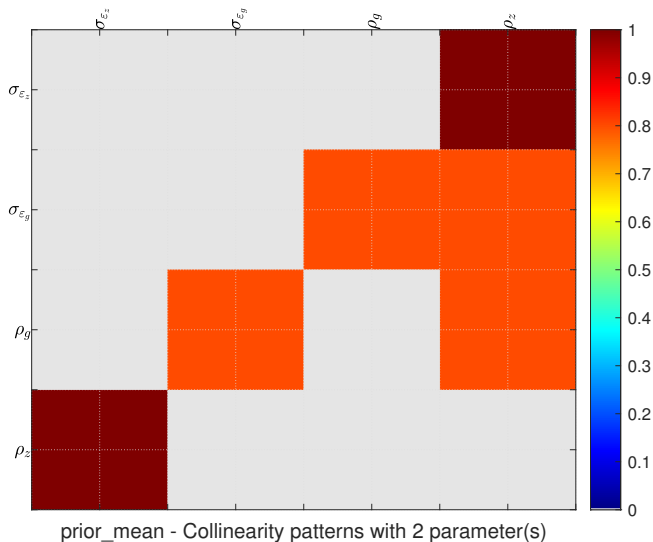
Dynare Output: collinearity patterns with 1 parameter

RBC_bayesian.mod



Dynare Output: collinearity patterns with 2 parameters

RBC_bayesian.mod

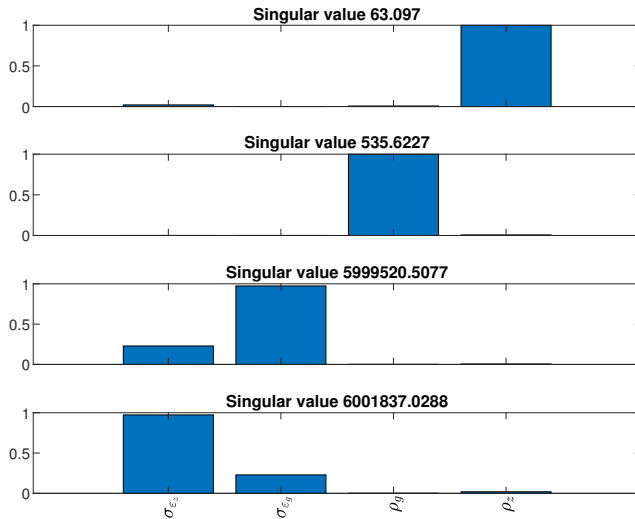


Singular Value Decomposition (SVD)

- Following Andrieu (2010), identification can also be judged from a singular value decomposition (SVD) of the information matrix
- Provides the size of the singular values and the associated eigenvectors (i.e. parameters)
- Parameter combinations associated with the smallest singular values are closest to being perfectly collinear and thus redundant
- Singular value of 0 implies that the parameter is completely unidentified as it is responsible for the information matrix being rank deficient

Dynare Output: SVD

RBC_bayesian.mod



Concluding remarks

- Fix obvious identification failures: calibrate or (better) re-parameterize your model
- Use readily available tools to get insight into the workings of your model
- Identifiability is model property that depends on choices a modeler makes
 - choice of observables
 - functional specifications
 - model features
 - choice of structural shocks
- Larger models tend to be theoretically identified, but suffer from weak identification
- Robust inference under possible weak identification
- wide confidence intervals for weakly identified parameters can be accompanied by narrow bands for IRFs

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